RESQ: EMERGENCY RESOURCES AND RESCUE MANAGEMENT SYSTEM

Micro Project Report

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By

SHADIYA S PRN25CS109
SHAHIDHA N PRN25CS110
SHALIN STEAPHEN PRN25CS111
SHARMI S PRN25CS112
PARVATHY S PRN25CS02

Guided by
ANISH A AZIZ
ASSISTANT PROFESSOR
COLLEGE OF ENGINEERING PERUMON

during
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
COLLEGE OF ENGINEERING PERUMON
CAPE - GOVT. OF KERALA, KOLLAM(D), KERALA

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College of Engineering
PERUMON

Introduction

RESQ (Emergency Resources and Rescue Management System) is a Java-based application developed to simplify emergency reporting and manage rescue operations effectively. Emergencies such as road accidents, fires, and other unexpected incidents require quick communication and proper coordination between the public and responsible authorities.

The system allows users to log in, report emergency incidents, view emergency details, and track the status of rescue operations. It also provides an administrative interface to allocate resources, manage available resources, and maintain the history of reported incidents.

The main aim of this project is to improve emergency response coordination, ensure proper resource utilization, and help authorities manage emergency situations more efficiently through a simple and user-friendly application.

Scope of the project

The scope of the RESCUE (Emergency Resources and Rescue Management System) is to provide an organized platform for reporting emergencies and managing rescue operations. The system allows users to report emergency incidents, view reported details, and track the progress of rescue activities.

It also provides administrators with the facility to allocate required resources, manage available resources, and maintain incident history. The application helps improve communication between users and authorities and supports better coordination during emergency situations.

In the future, the system can be enhanced with features such as real-time location tracking, GPS integration, instant notifications, and direct communication with emergency services. This will make emergency response faster and more effective.

Technology used

The RESCUE (Emergency Resources and Rescue Management System) is developed using the following technologies:

1. Java: Java is used as the main programming language to develop the application and implement its functionalities.
2. Java Swing: Java Swing is used to design the graphical user interface (GUI), including login forms, dashboards, buttons, text fields, and other components.
3. AWT (Abstract Window Toolkit): AWT is used for GUI layout management, fonts, colors, and event handling.
4. JFrame: JFrame is used to create the main application windows and different frames.
5. Event Handling: Java event handling is used to perform actions such as login, submitting emergency reports, allocating resources, updating incident status, and tracking emergencies.
6. Object-Oriented Programming (OOP): Classes and objects are used to organize the application into separate modules and manage emergency and resource information.
7. Java Collections: ArrayList and HashMap are used to store and manage emergency reports and resource details during application execution.

Problem statement

During emergency situations such as road accidents, fires, and other unexpected incidents, reporting emergencies and coordinating rescue operations can be difficult. Lack of proper communication, delays in reporting, and improper resource allocation may affect the response process. There is a need for a simple and organized system to report emergencies and manage rescue activities efficiently.

The RESCUE system aims to solve these problems by providing a platform where users can report emergency incidents, view incident details, and track the status of rescue operations. It also helps administrators allocate and manage resources and maintain incident history. The system is developed to improve communication, coordination, and emergency management.

Literature Review / Background

Emergency management systems have become increasingly important with the growth of information technology. Several existing systems and studies focus on improving emergency reporting, disaster management, and resource allocation. Traditional emergency response methods mainly depend on phone calls and manual communication, which can lead to delays and difficulties in managing emergency situations.

Existing emergency management applications provide facilities for reporting incidents, storing emergency details, and coordinating rescue activities. These systems help authorities receive information quickly and make better decisions during emergencies. Resource management systems also play an important role in ensuring the proper allocation and utilization of available resources.

The RESCUE project is developed based on these concepts. It combines emergency reporting, incident information management, response tracking, and resource allocation in a single Java-based application. Unlike manual methods, the system provides a structured way to manage emergency information and supports better coordination between users and administrators.

Implementation

MODULES OF THE RESOURCES AND RESCUE MANAGEMENT SYSTEM

The RESQ (Emergency Resources and Rescue Management System) is implemented using Java and Java Swing. The application provides a graphical user interface through which users can report emergencies and administrators can manage rescue resources.

1. Login Module

The Login Frame allows users and administrators to access the application. After successful login, users are directed to the User Dashboard and administrators are directed to the Admin Dashboard

2.User Dashboard

The User Dashboard provides access to the main user functions. Users can report a new emergency, view emergency details, and track the response status of reported incidents.

3. Emergency Report Frame

This frame allows users to submit emergency information by selecting the emergency type, priority, location, number of people affected, and required emergency service. The system validates the entered details, generates a unique Emergency ID, and stores the report.

4. Emergency Details Module

This frame displays the details of a submitted emergency, including Emergency ID, emergency type, priority, location, number of people affected, required service, and current status.

5. Response Tracking Module

This frame allows users to track an emergency using its Emergency ID. It displays the emergency details, assigned resources, allocated quantity, and current response status.

6. Admin Dashboard

The Admin Dashboard provides administrators with access to emergency and resource management functions. It displays the total number of reports and the

number of pending, active, and resolved emergencies. It also shows the latest emergency reports.

7. Resource Allocation

This frame allows administrators to allocate resources such as Police, Fire Force, Ambulance, and Medical Kits to reported emergencies. The system checks resource availability and validates the allocation quantity before updating the emergency status.

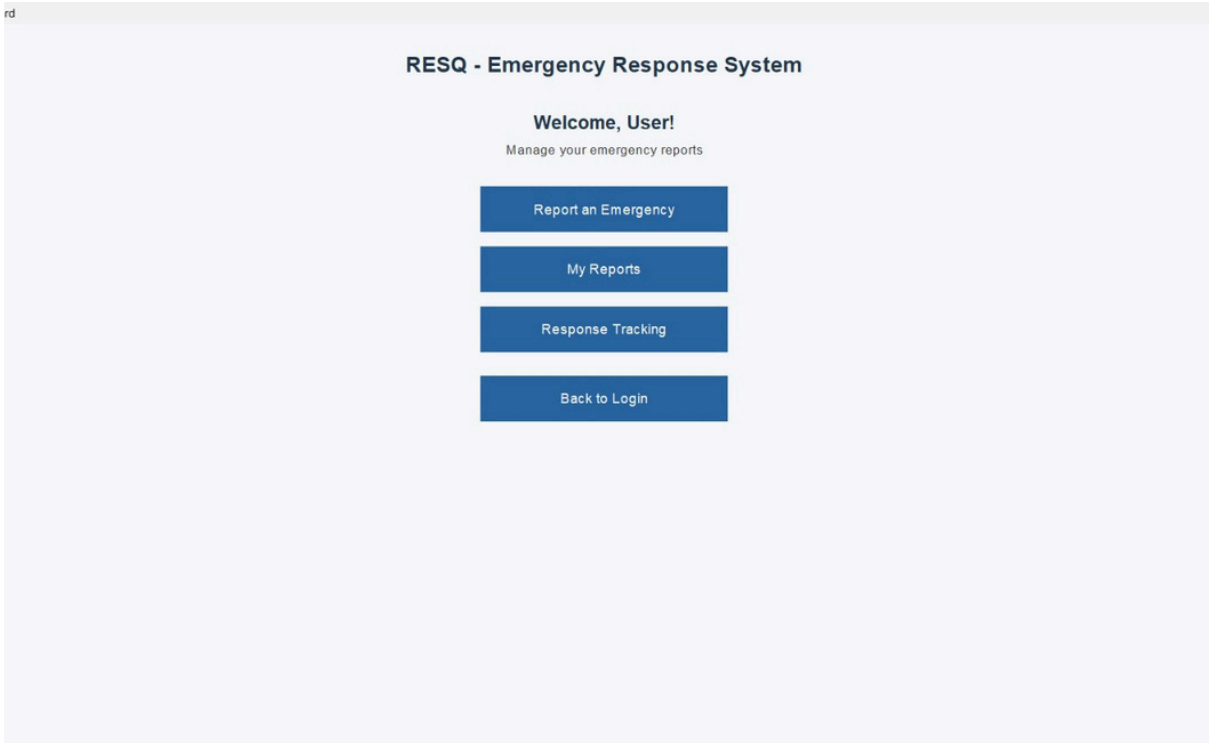
8. Resource Management

This frame allows administrators to manage the available quantity of different emergency resources. It displays resource availability and updates the resource stock while ensuring that the total quantity does not fall below the already allocated quantity.

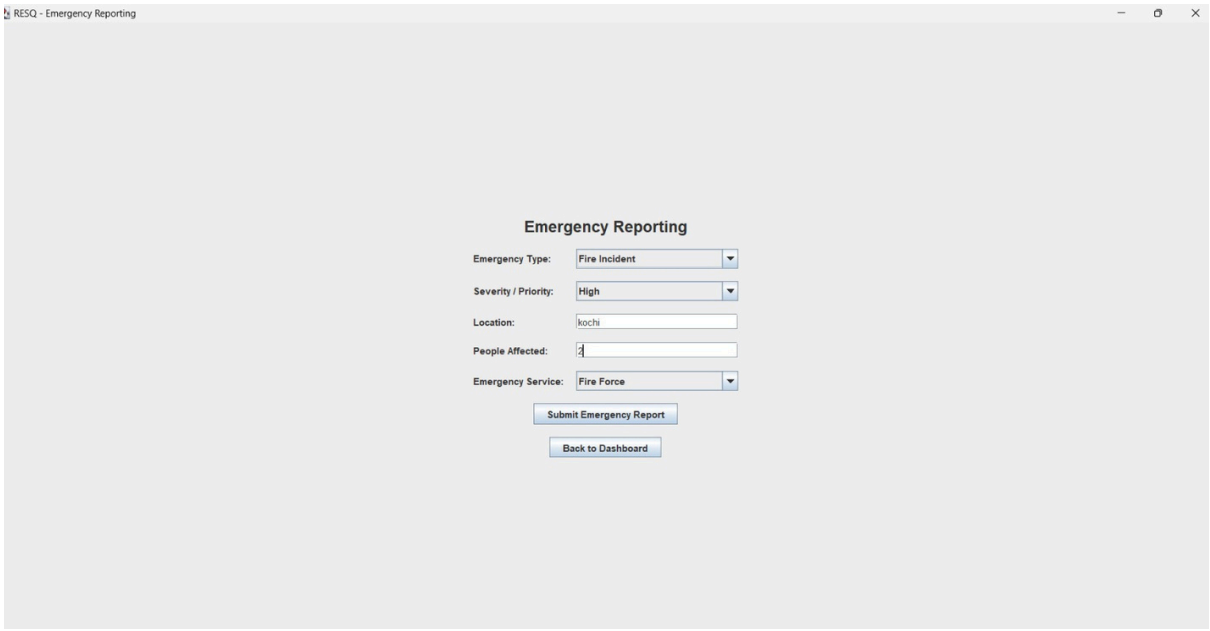
9. Incident History

This frame allows administrators to search for incidents using the Incident ID, view incident details, and update the status of emergencies. The available statuses include Reported, Pending, Resources Allocated, Active, and Resolved.

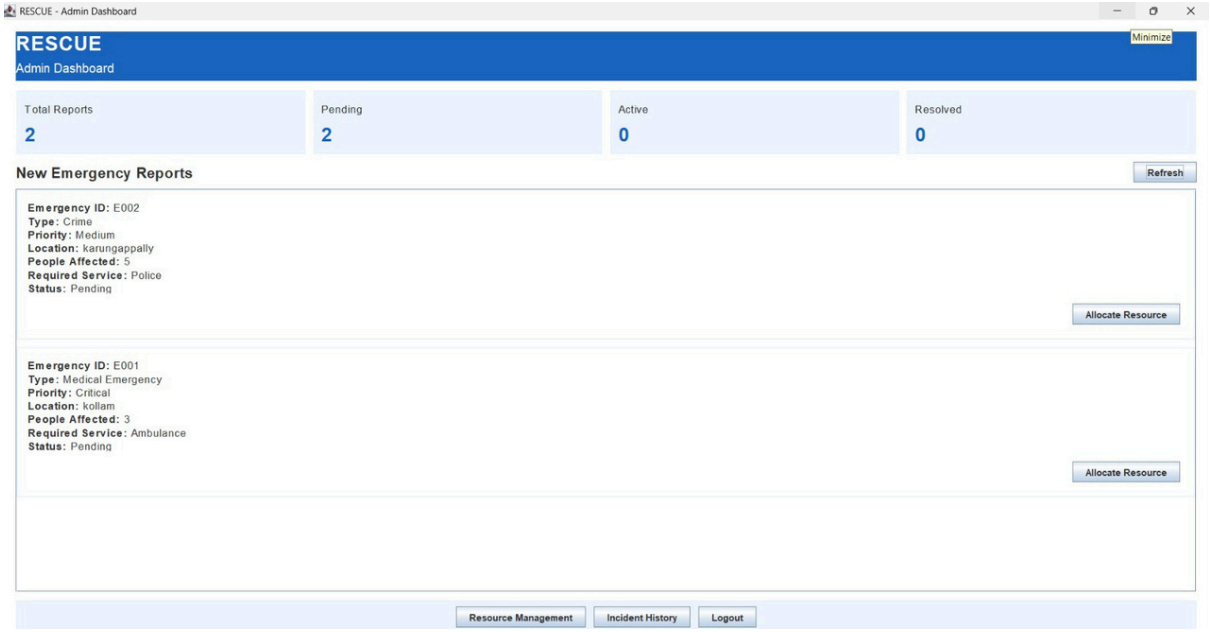
User Dashboard Frame



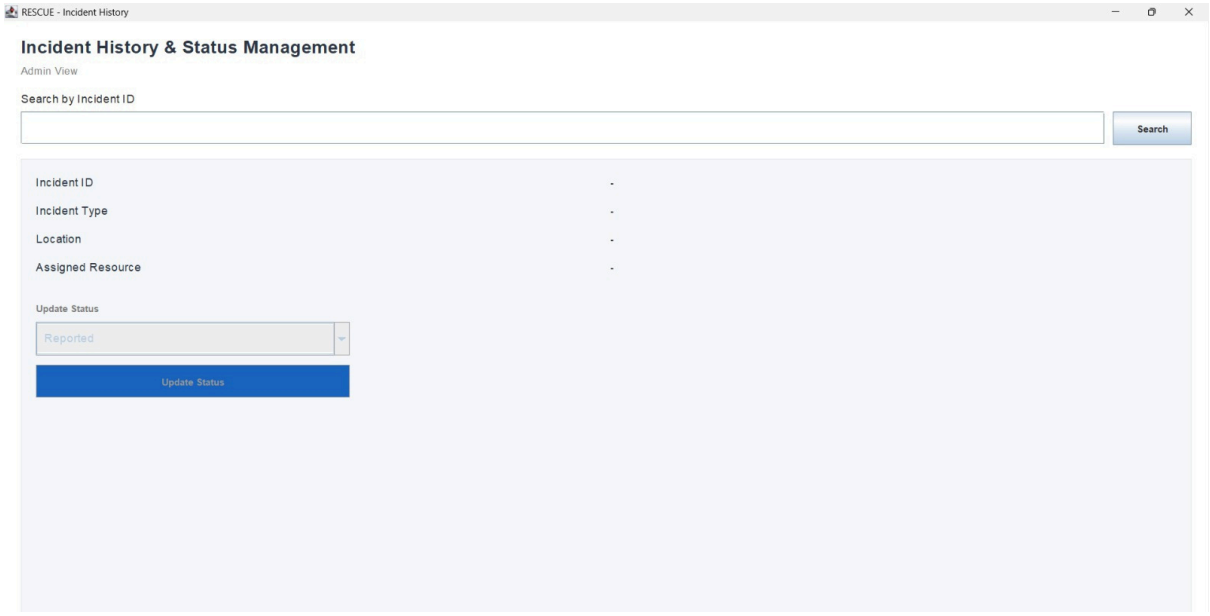
Emergency Reporting



Admin Dashboard



Incident History&Status updation

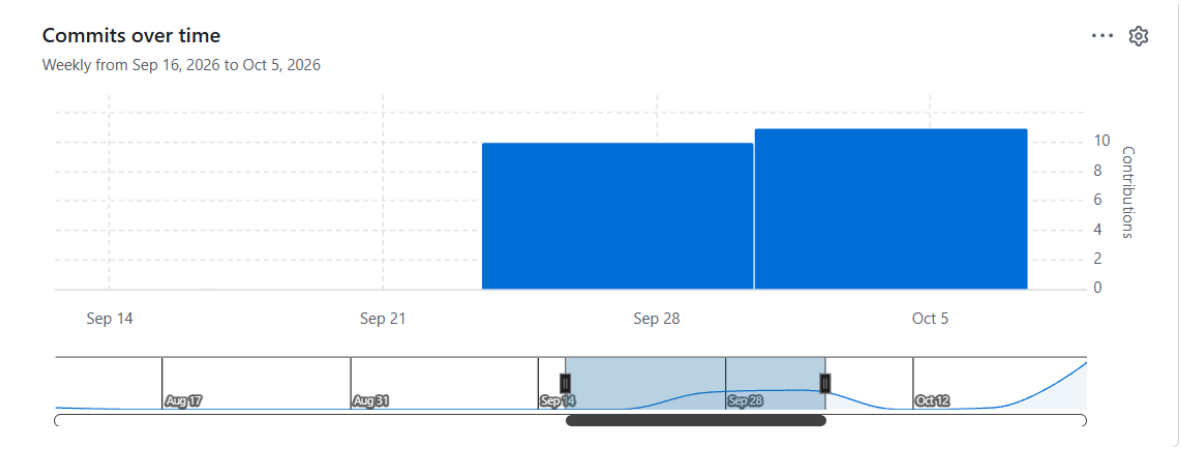


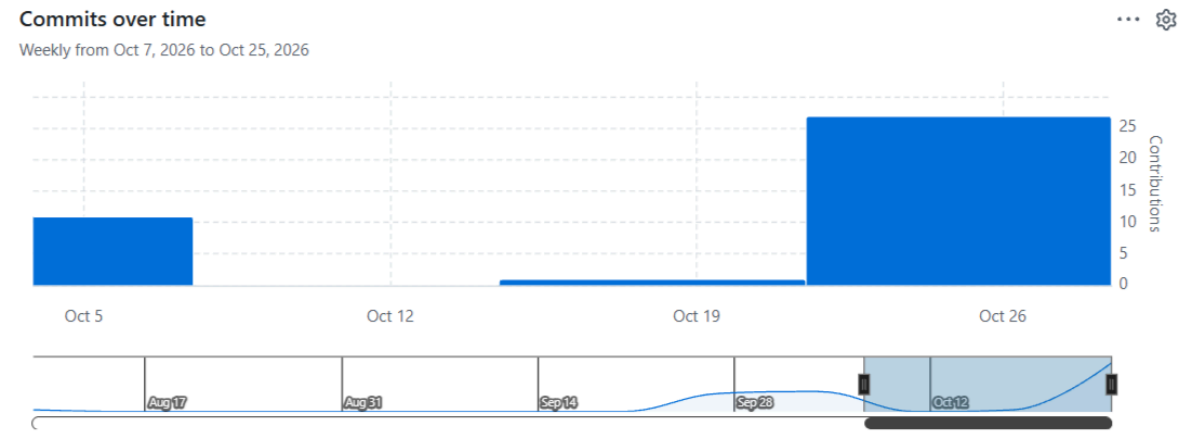
Results and Discussion

The RESCUE (Emergency Resources and Rescue Management System) was successfully developed using Java and Java Swing. It provides separate interfaces for users and administrators to report emergencies, view incident details, track rescue operations, and manage resources.

The project demonstrates the practical application of object-oriented programming, GUI development, event handling, and exception handling in Java. GitHub was used for collaborative development and version control. Team members contributed to the project through individual commits, and the commit history provides a record of code updates and development progress.

Overall, the system simplifies emergency reporting and resource management through an organized and user-friendly application.





Conclusion

The RESCUE (Emergency Resources and Rescue Management System) is a Java-based application developed to simplify emergency reporting and rescue management. The system allows users to report emergencies, view incident details, and track the status of rescue operations. It also helps administrators allocate resources, manage available resources, and maintain incident history.

This project provides a simple and organized way to manage emergency information and improve coordination between users and authorities. It helps ensure better resource utilization and supports the effective management of emergency situations. Thus, the RESCUE system serves as a useful application for emergency reporting and rescue management.

GitHub path

https://github.com/sharmisanofar9567-jpg/java.git

Reference

1. Oracle, Java Documentation. https://docs.oracle.com/en/java/

2. Oracle, The Java Swing Tutorial.
https://docs.oracle.com/javase/tutorial/uiswing/

3. Oracle, Java SE API Documentation.
https://docs.oracle.com/en/java/javase/25/docs/api/

4. GitHub, RESQ:Emergency Resources and Rescue Management System – Project Source Code. https://github.com/sharmisanofar9567-jpg/java